RetailPulse – AI-Powered Customer Analytics & Demand Forecasting Platform

Zidio Development – Data Science & Analytics Domain

Internship Project

Developed By:

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Project Duration: 1 Month (20 September – 20 October 2026)

1: Executive Summary & Project Overview

1.1 Executive Summary

RetailPulse is an advanced, AI-powered retail analytics platform designed to help businesses understand customer behavior, forecast product demand, and optimize inventory management through machine learning and MLOps practices. The project bridges the gap between raw multi-source retail data and data-driven executive decision-making.

1.2 Key Highlights & Scope

- **Customer Segmentation:** Leverages behavioral and RFM (Recency, Frequency, Monetary) metrics via K-Means clustering (4 to 6 clusters) to group buyers into actionable segments (e.g., High-Value Frequent Buyers, Active Shoppers, and At-Risk targets).
- **Advanced Demand Forecasting:** Employs regression and time-series models (including Baseline, Weekday/Weekend, Seasonal, and Random Forest Regressors) to generate robust 30-day ahead sales predictions.
- **Churn Prediction & Risk Mitigation:** Utilizes powerful classification algorithms (Logistic Regression and XGBoost) coupled with SHAP explainability to identify customer inactivity risks proactively.
- **Inventory Optimization:** Translates forecasted demand directly into intelligent inventory recommendations, safety stock levels, and reorder thresholds (inventory_clean.csv) to minimize overstocking and stockout risks.
- **Enterprise Business Intelligence:** Powered exclusively by **Power BI** (RetailPulse.pbix), delivering real-time visual insights, exploratory data analysis metrics, and dynamic reporting layers.

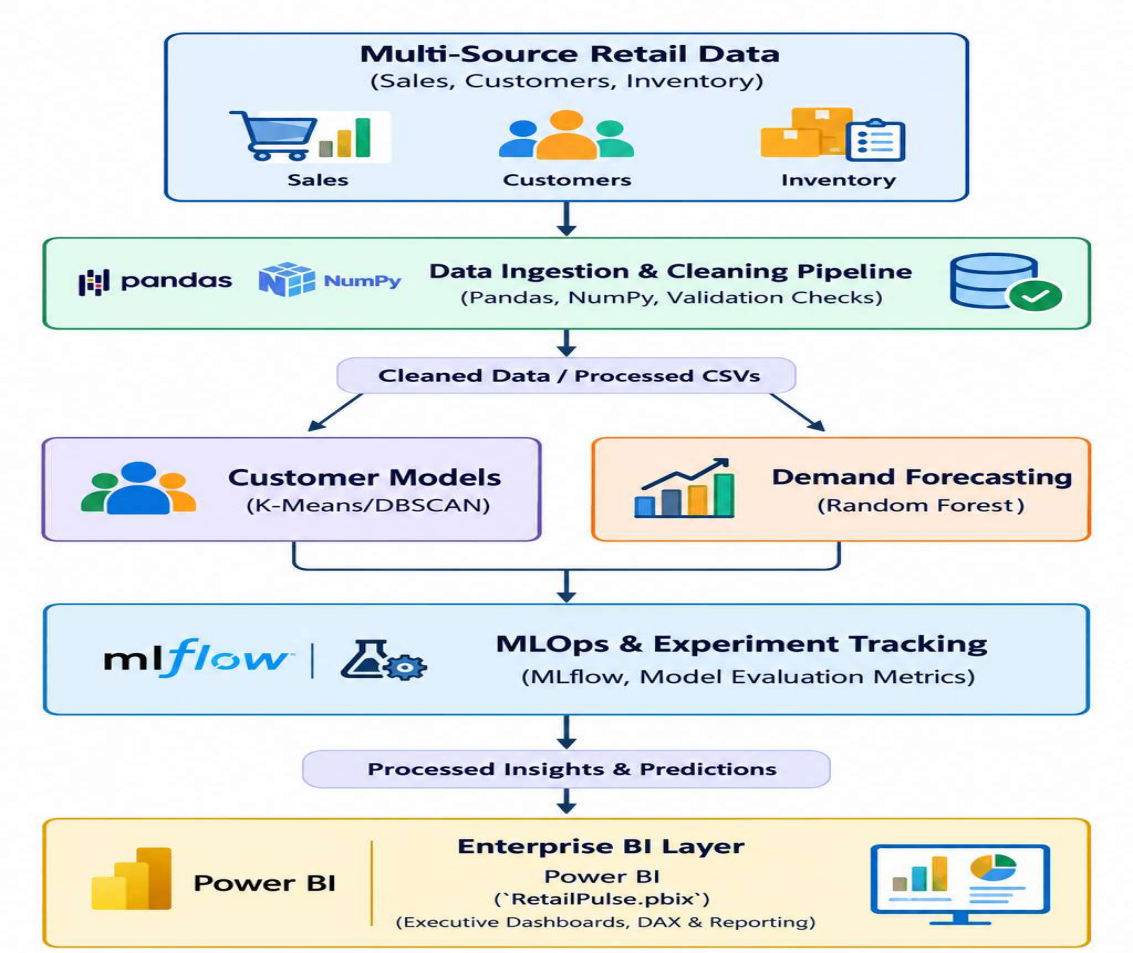
2: System Architecture & Data Pipeline

The platform is built on a modular, enterprise-ready architecture structured across four core operational layers:

1. **Data Ingestion & Storage Layer:** Raw sales transactions (sales_daily.csv), inventory snapshots (inventory_snapshots.csv), and calendar logs (calendar.csv) are ingested and processed using Python (Pandas/NumPy) into standardized CSV datasets (sales_clean.csv, inventory_clean.csv, calendar_clean.csv).
2. **Analytics & Machine Learning Layer:** Cleaned data is fed into specialized predictive modules for customer clustering, demand forecasting, and inventory threshold calculation.
3. **MLflow MLOps Layer:** Tracks model training runs, versions hyperparameters, and monitors performance metrics and data drift.

4. **Power BI Business Intelligence Layer (RetailPulse.pbix):** Connects directly to processed datasets to serve deep executive dashboards, DAX-calculated KPIs, and advanced data modeling.

High-Level Architecture Flow Diagram



Technology Stack

- **Programming Language:** Python
- **Data Processing & Machine Learning:** Pandas, NumPy, Scikit-learn, aur XGBoost
- **Forecasting Models:** Random Forest, Prophet, aur Time-Series Analysis
- **Experiment Tracking & MLOps:** MLflow
- **Business Intelligence & Dashboard:** Power BI
- **Version Control & Hosting:** GitHub

3: Data Exploration & Profiling (01_data_exploration.ipynb)

The project began with deep exploratory data analysis across three core datasets:

- **Sales Data Profiling:** Analyzed 36,550 transaction records across 50 unique SKUs, validating price, revenue, and promotion distributions (df.shape, df.info(), df.describe()).

- **Inventory Data Profiling:** Assessed 4,800 inventory snapshot records across 200 unique SKUs, verifying current stock, lead times, safety stock, and inventory valuations.
- **Calendar Profiling:** Examined 731 date entries (2024–2025) covering seasonal attributes, holiday indicators, and promotion events.

Sales Dataset Structure and Head Preview

```
Libraries imported successfully!
Files available: ['calendar.csv', 'inventory_snapshots.csv', 'sales_daily.csv']
```

	Date	SKU	Units_Sold	Revenue	Price	Promotion
0	2024-01-01	SKU001	5	18320.25	3664.05	0
1	2024-01-01	SKU002	15	57085.35	3805.69	0
2	2024-01-01	SKU003	5	40391.15	8078.23	0
3	2024-01-01	SKU004	5	34307.85	6861.57	0
4	2024-01-01	SKU005	12	113918.64	9493.22	0

Numerical Summary and Descriptive Statistics of Sales Data

	Units_Sold	Revenue	Price	Promotion
count	36550.000000	36550.00000	36550.000000	36550.000000
mean	14.003010	84707.43385	5892.663800	0.102599
std	8.467305	76534.63044	3138.775609	0.303439
min	0.000000	0.00000	663.460000	0.000000
25%	7.000000	27643.70000	3484.090000	0.000000
50%	13.000000	56125.90000	5649.795000	0.000000
75%	20.000000	127954.32000	8516.600000	0.000000
max	56.000000	546977.00000	11642.450000	1.000000

4: Data Cleaning & Preprocessing (02_data_cleaning.ipynb)

To ensure high data integrity:

- **Missing Value Handling:** Missing calendar holiday labels and promotion events were systematically handled by filling them with "No Holiday" and "No Promotion".
- **Standardization:** Date columns were standardized to datetime format across all tables.
- **Dataset Integration:** Sales data was merged with calendar attributes using a validated many-to-one relationship, producing the integrated file sales_calendar_clean.csv (36,550 rows, 16 columns with 0 missing values and 0 duplicates).

5: Exploratory Data Analysis & Trends (01 & 03)

Exploratory scripts and visualization routines uncovered clear operational trends:

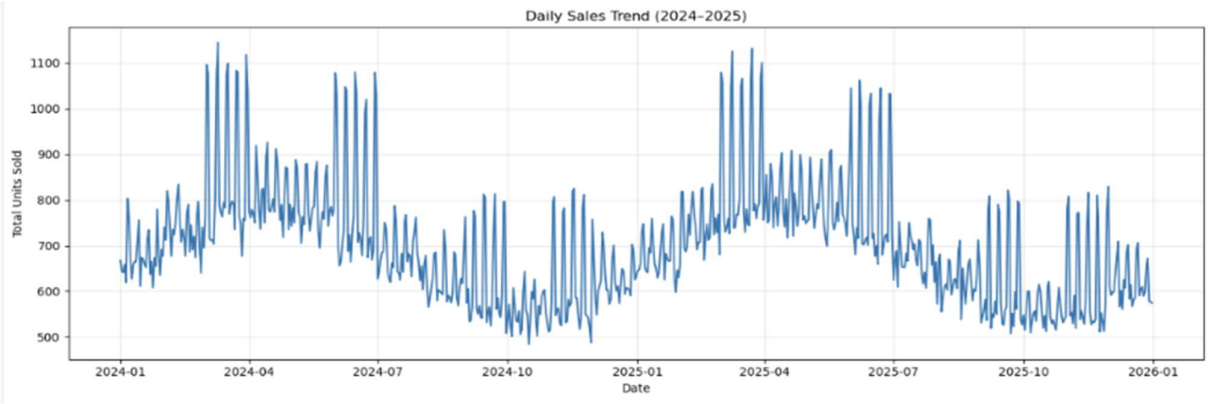
- **Day-of-Week Patterns:** Weekend sales averaged significantly higher (~818 units on Sundays, ~813 on Saturdays) compared to weekdays (~651 units).

- **Monthly Revenue Trends:** Analyzed total monthly revenue progression across the 2-year timeline (peaking dynamically during seasonal promotional windows).

Monthly Revenue Trend (2024–2025)



Daily Sales Trend (2024–2025)

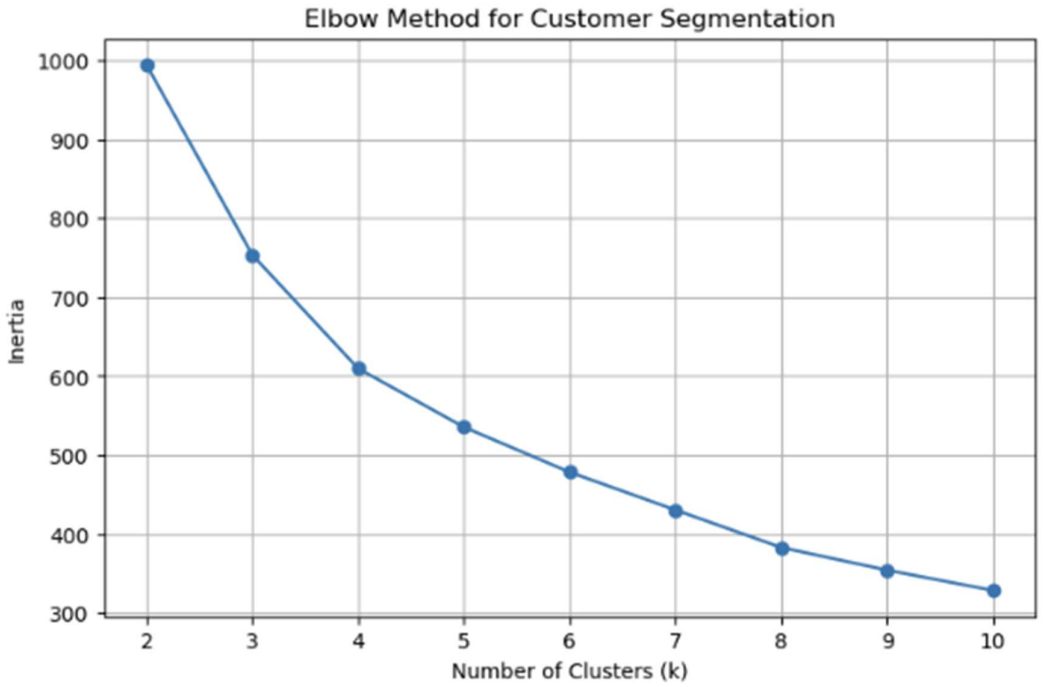


6: Advanced Machine Learning & Customer Segmentation (05_customer_segmentation.ipynb)

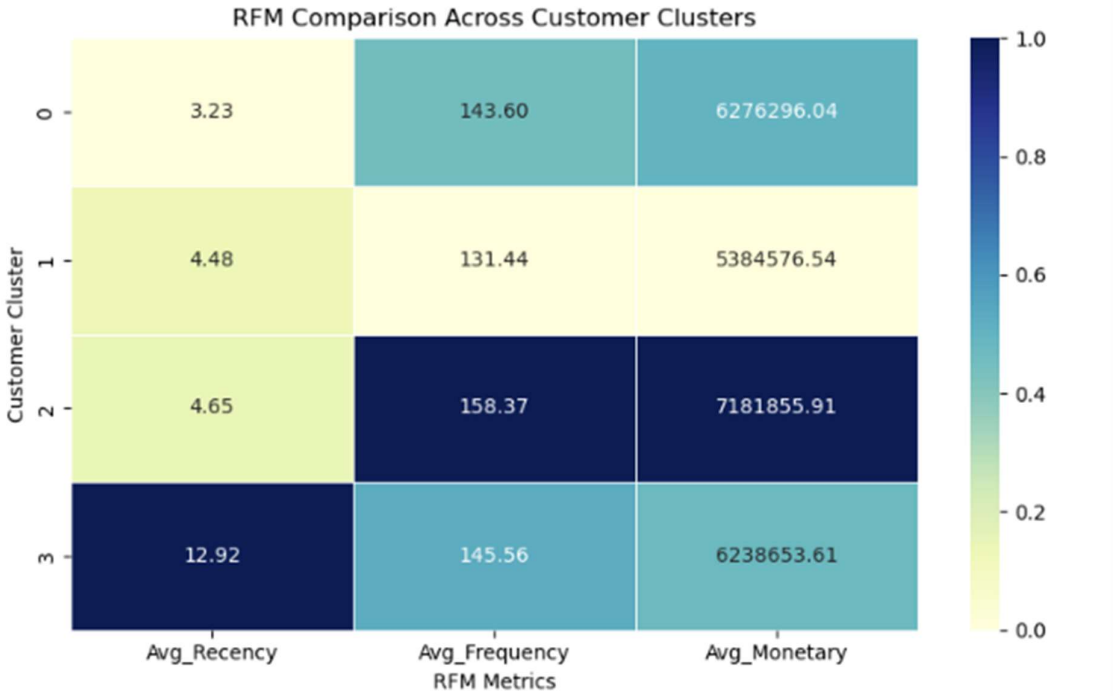
Customer behavior was segmented using RFM metrics computed from customer transaction histories (500 unique customers):

- **Feature Scaling:** Applied StandardScaler on Recency, Frequency, and Monetary metrics.
- **K-Means Clustering:** Determined optimal cluster counts using the Elbow Method and inertia plots. Customers were segmented into distinct behavioral groups (e.g., High-Value Frequent Buyers, Recent Active Customers, and Re-engagement Targets).
- **DBSCAN Clustering:** Explored density-based clustering to detect noise points and outlier shopping behaviors.

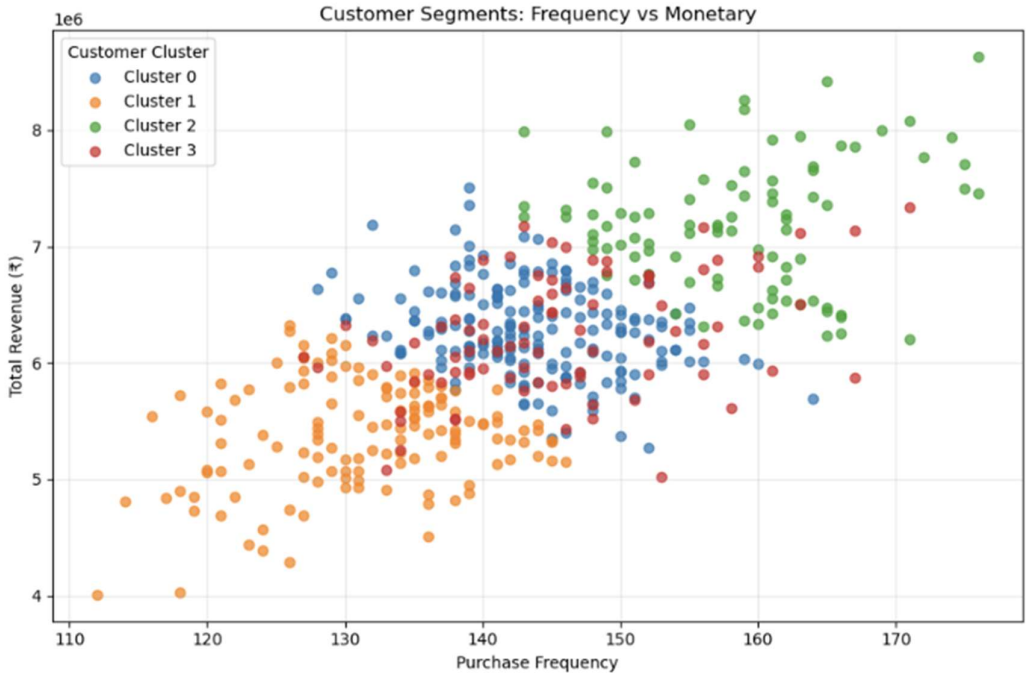
Elbow Method Plot for Customer Segmentation



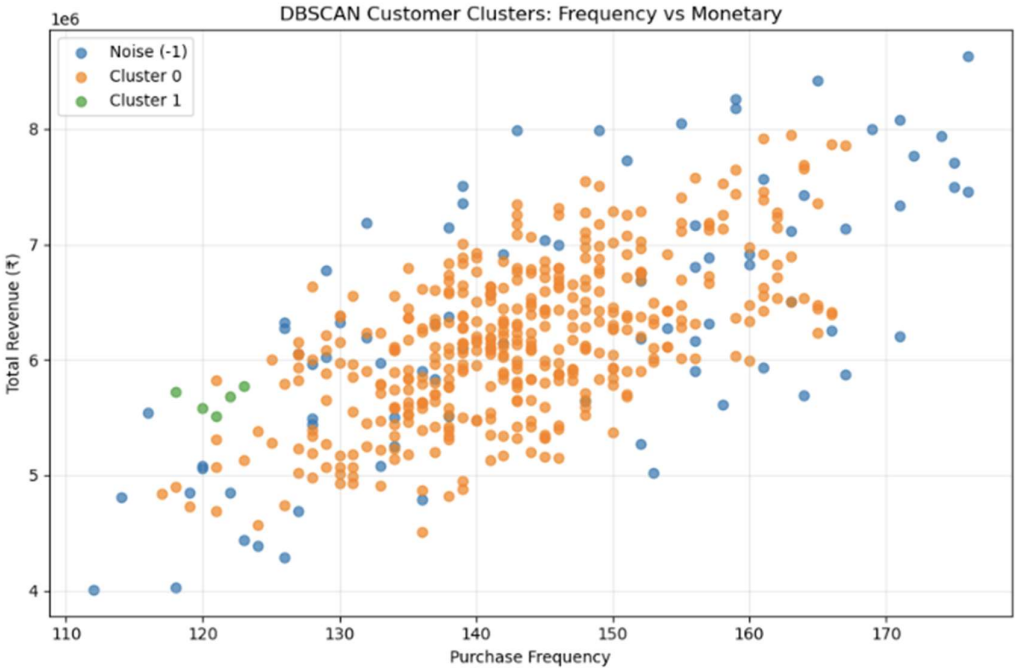
Cluster Heatmap



Customer Segments Frequency VS Monetary clusters chart



DBSCAN Customer Clusters chart

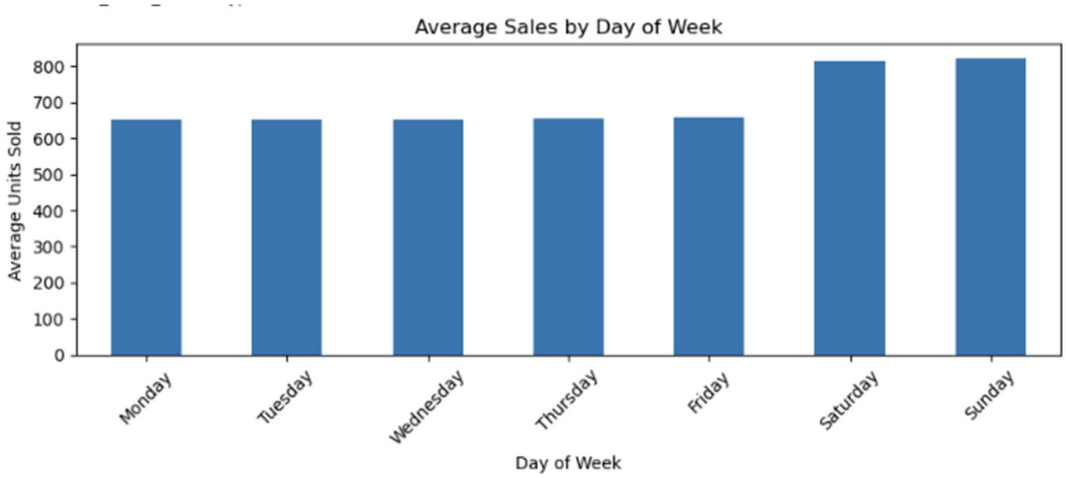


7: Demand Forecasting & Predictive Modeling (03_demand_forecasting.ipynb)

Daily total units sold across the retail enterprise were modeled to generate accurate 30-day ahead projections:

- **Chronological Train-Test Split:** Trained on historical daily sales from Jan 2024 to June 2025 and tested on July 2025 to Dec 2025.
- **Baseline & Weekday/Weekend Models:** Evaluated historical means and day-type averages (achieving a baseline test MAE of 122.1 and Weekday/Weekend MAE of 109.29).
- **Random Forest & Regression Models:** Implemented advanced regressors incorporating rolling averages and calendar indicators to capture seasonal demand spikes.

Average Sales by Day of Week Bar Chart

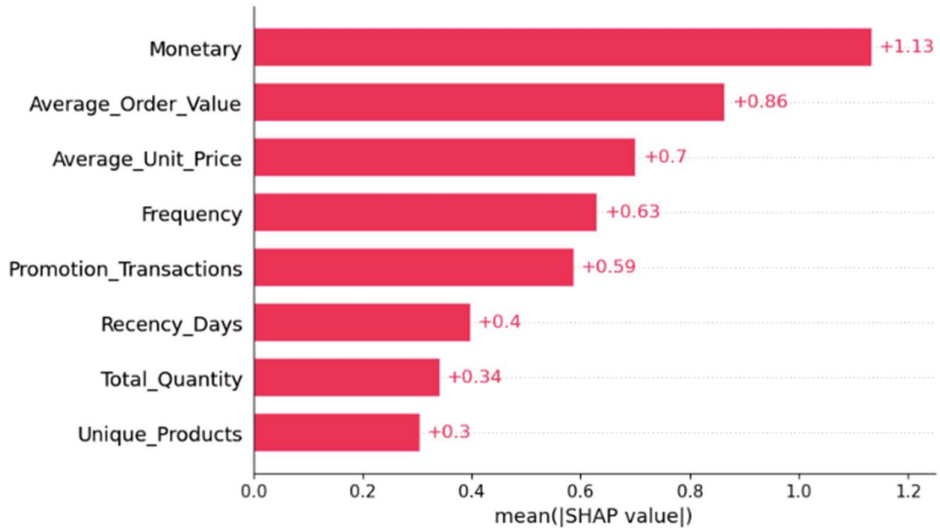


8: Churn Prediction & Model Explainability (06_churn_prediction.ipynb)

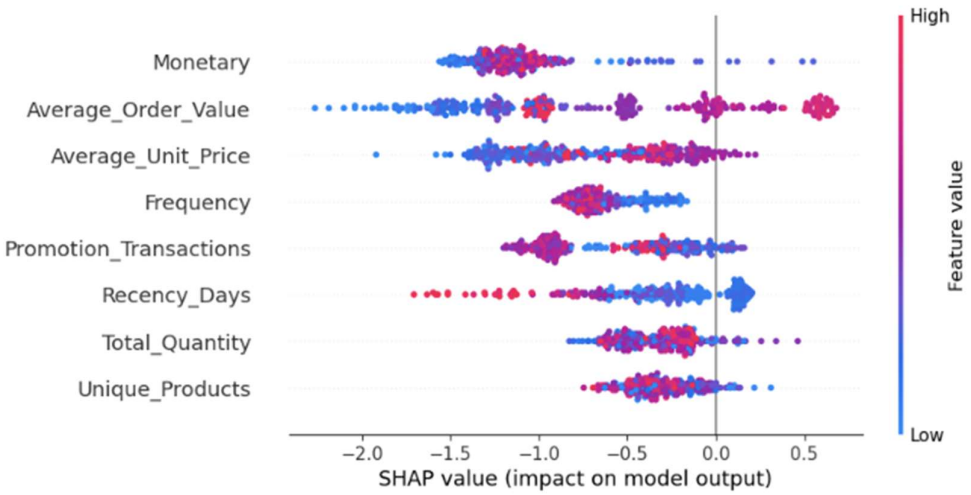
Customer retention and inactivity risks were evaluated:

- **Feature Engineering:** Extracted customer-level metrics including purchase frequency, monetary spend, average order value, and recency days.
- **Classification Modeling:** Tested Logistic Regression and XGBoost classifiers to flag inactive or at-risk customers.
- **Model Explainability (SHAP):** Integrated SHapley Additive exPlanations (shap.plots.bar / beeswarm plots) to interpret feature importance and transparency for stakeholders.

Plot global feature importance using SHAP values



Show how features influence churn predictions

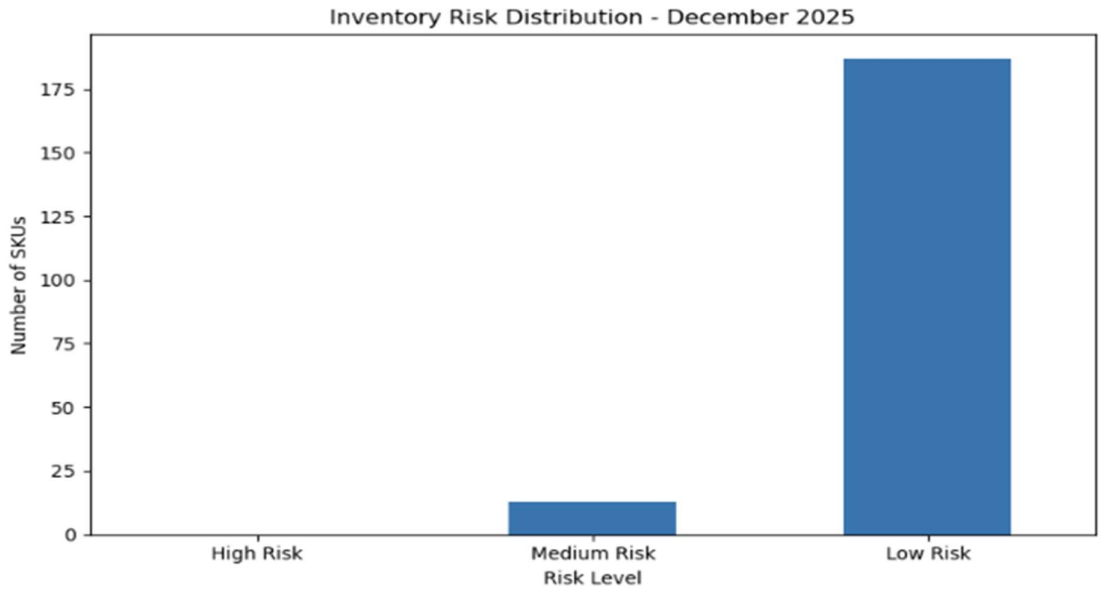


9: Inventory Optimization & Reorder Thresholds (04_inventory_analysis.ipynb)

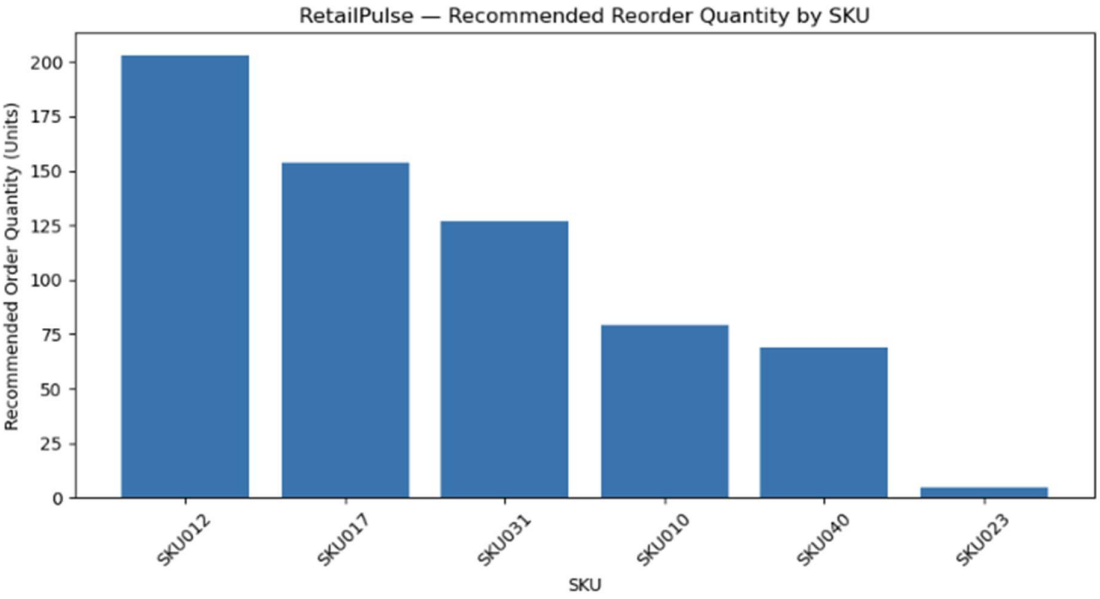
Inventory health and replenishment needs were computed using inventory_clean.csv and 30-day sales forecasts (sales_forecast_next_30_days.csv):

- **Reorder Point Calculation:** Evaluated lead times, safety stock thresholds, and current stock vs. on-order quantities.
- **Risk Categorization:** Identified SKUs requiring immediate reordering (Reorder Needed vs. Stock Sufficient) and generated precise recommended order quantities.

Inventory Risk Distribution Bar Chart



Recommended Reorder Quantity by SKU Bar Chart

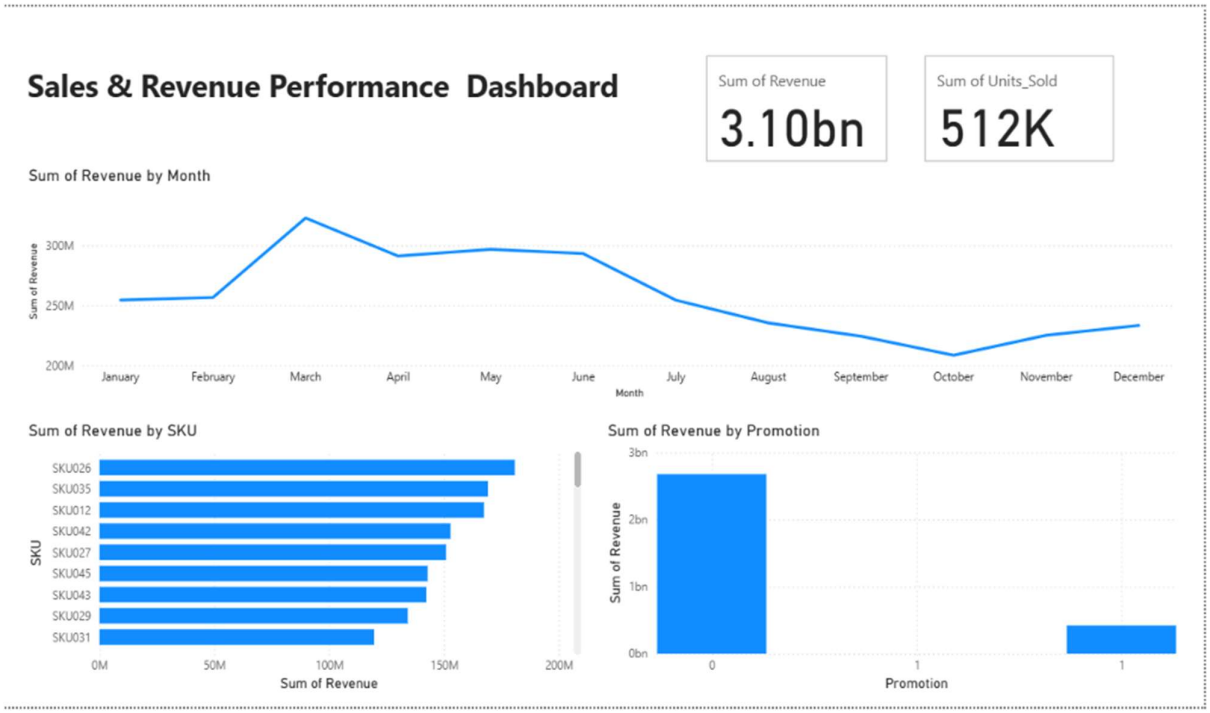


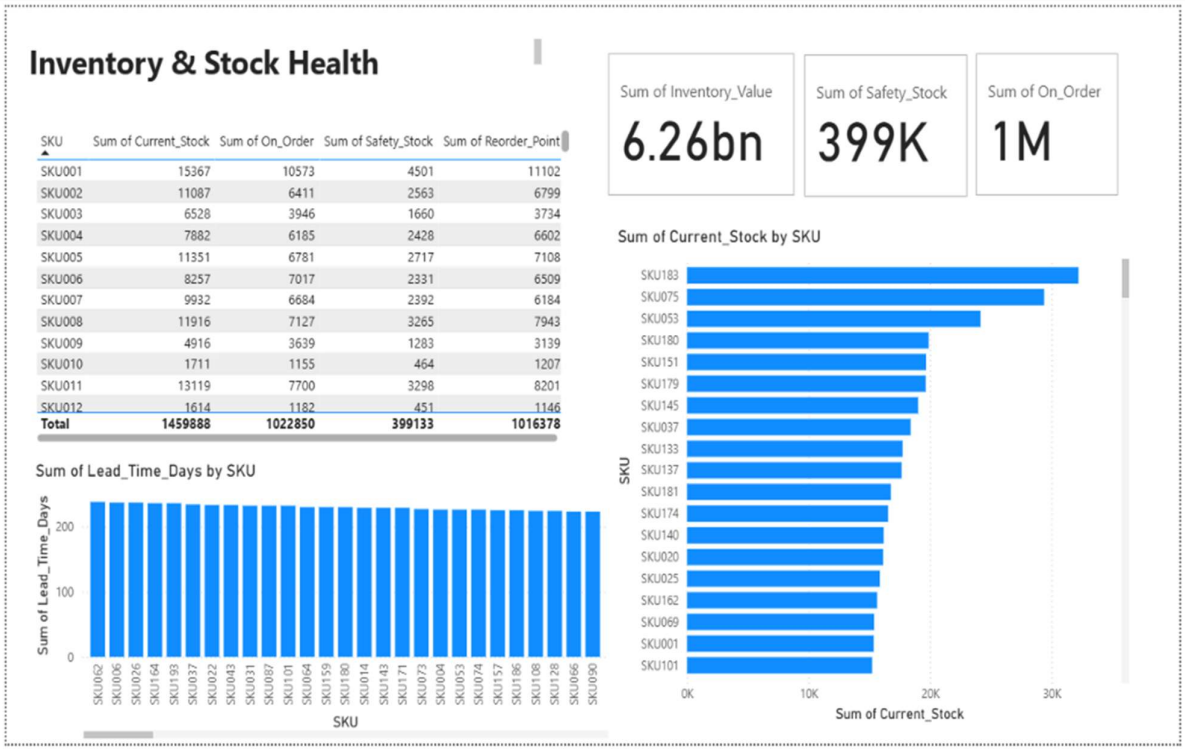
10: Enterprise Business Intelligence, Best Practices & Reflection

10.1 Enterprise Power BI Dashboard (RetailPulse.pbix)

All processed datasets (sales_calendar_clean.csv, inventory_clean.csv, customer_segments.csv) were imported into **Power BI** to deliver real-time executive reporting, DAX-calculated KPIs, and interactive visual dashboards.

Dashboard Screenshot :





Customer Segmentation Dashboard

Count of Customer_ID
500

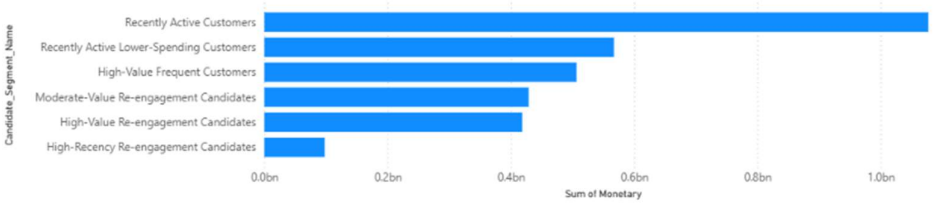
Count of Candidate_Cluster
6

Sum of Frequency
72K

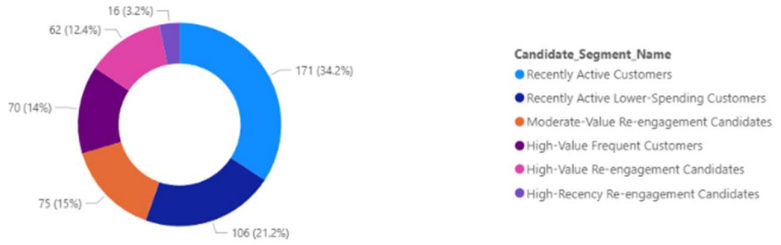
Sum of Candidate_Cluster
995

Sum of Recency
3K

Sum of Monetary by Candidate_Segment_Name



Count of Customer_ID by Candidate_Segment_Name



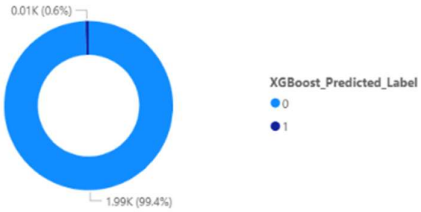
Churn Prediction & Risk

Count of Customer_ID
10

Average of ROC_AUC
0.57

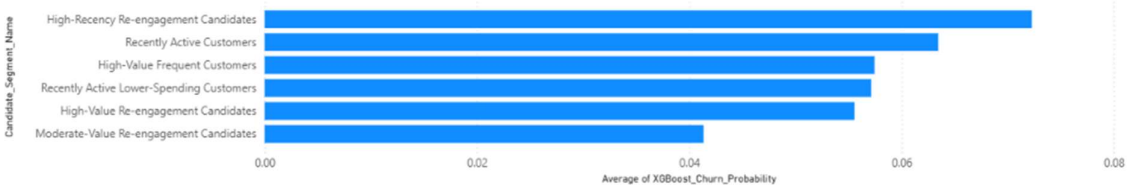
Sum of Precision_At_Top_20_Percent
0.01

Count of Customer_ID by XGBoost_Predicted_Label



Customer_ID	Average of XGBoost_Churn_Probability	Average of XGBoost_Predicted_Label
CUST0006	0.58	1.00
CUST0190	0.68	1.00
CUST0196	0.61	1.00
CUST0224	0.57	1.00
CUST0305	0.52	1.00
CUST0320	0.56	1.00
CUST0327	0.68	1.00
CUST0417	0.54	1.00
CUST0432	0.65	1.00
CUST0453	0.53	1.00
Total	**0.60**	**1.00**

Average of XGBoost_Churn_Probability by Candidate_Segment_Name



10.2 Deployment :

- **Version Control (GitHub):** We have hosted and managed the entire project's code, notebooks, and project files by creating a repository on GitHub for tracking.
- **Cloud Integration (AWS):** AWS (Amazon Web Services) has been used for data warehousing and secure storage, from where all raw and processed data is fetched.
- **BI Deployment (Power BI):** The final dashboards have been deployed on Power BI Service, which connect directly to the AWS data source via a live connection and provide real-time insights.

10.3 Industry Best Practices Applied

- **Modular Code Architecture:** Clean workspace separation for raw data, processed datasets, and analytical notebooks (01 to 06).
- **Data Integrity Checks:** Rigorous cross-dataset consistency verification and business rule validation.
- **Version Control & MLOps:** Code management via Git & GitHub, and experiment tracking via MLflow.

10.4 Personal Reflection & Future Roadmap

- **Core Insight:** Successfully bridged raw multi-source retail data with enterprise business intelligence reporting.
- **Future Roadmap:** Integrating live streaming data pipelines for real-time updates, implementing deep learning neural architectures for SKU-level forecasting, and expanding Power BI cloud gateway automation.